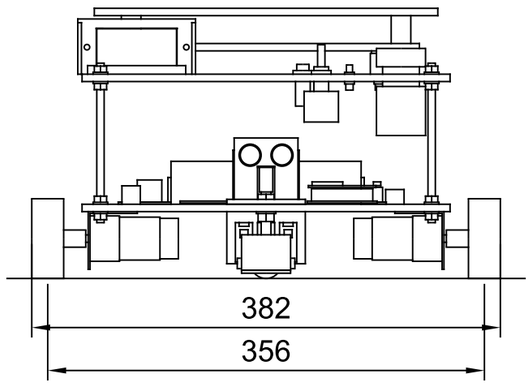
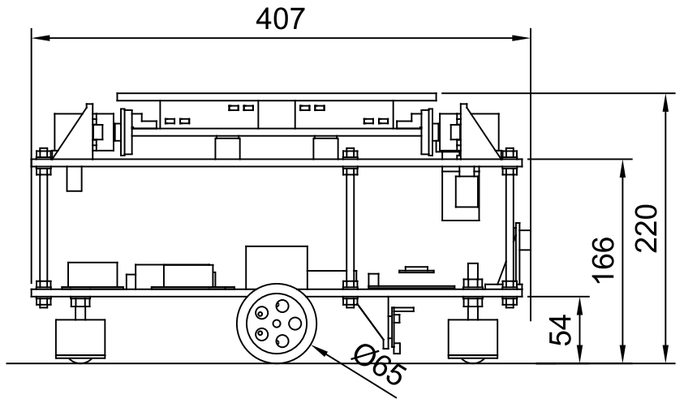
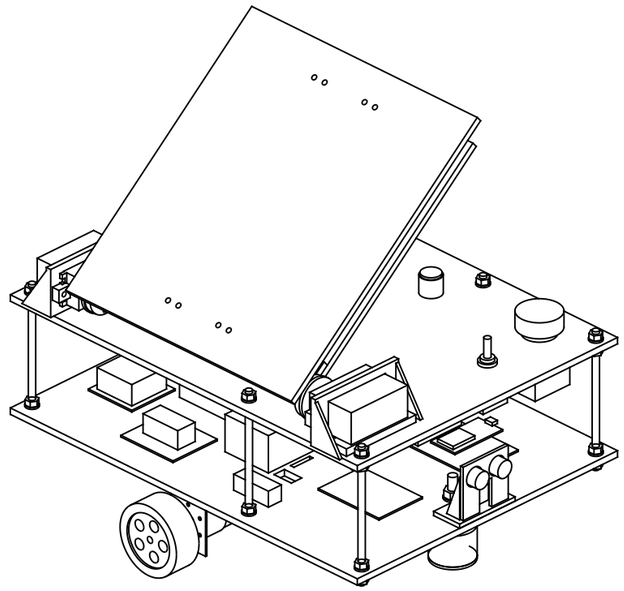


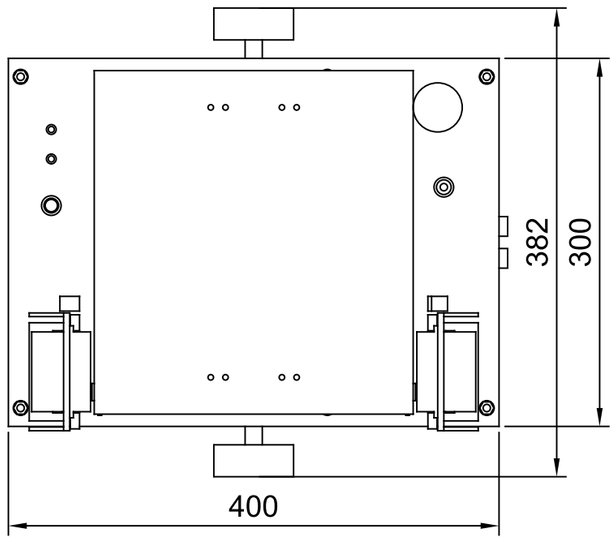
FRONT VIEW



SIDE VIEW (from the right) FRONT -->



ISOMETRIC VIEW - BED TIPPED 40 deg

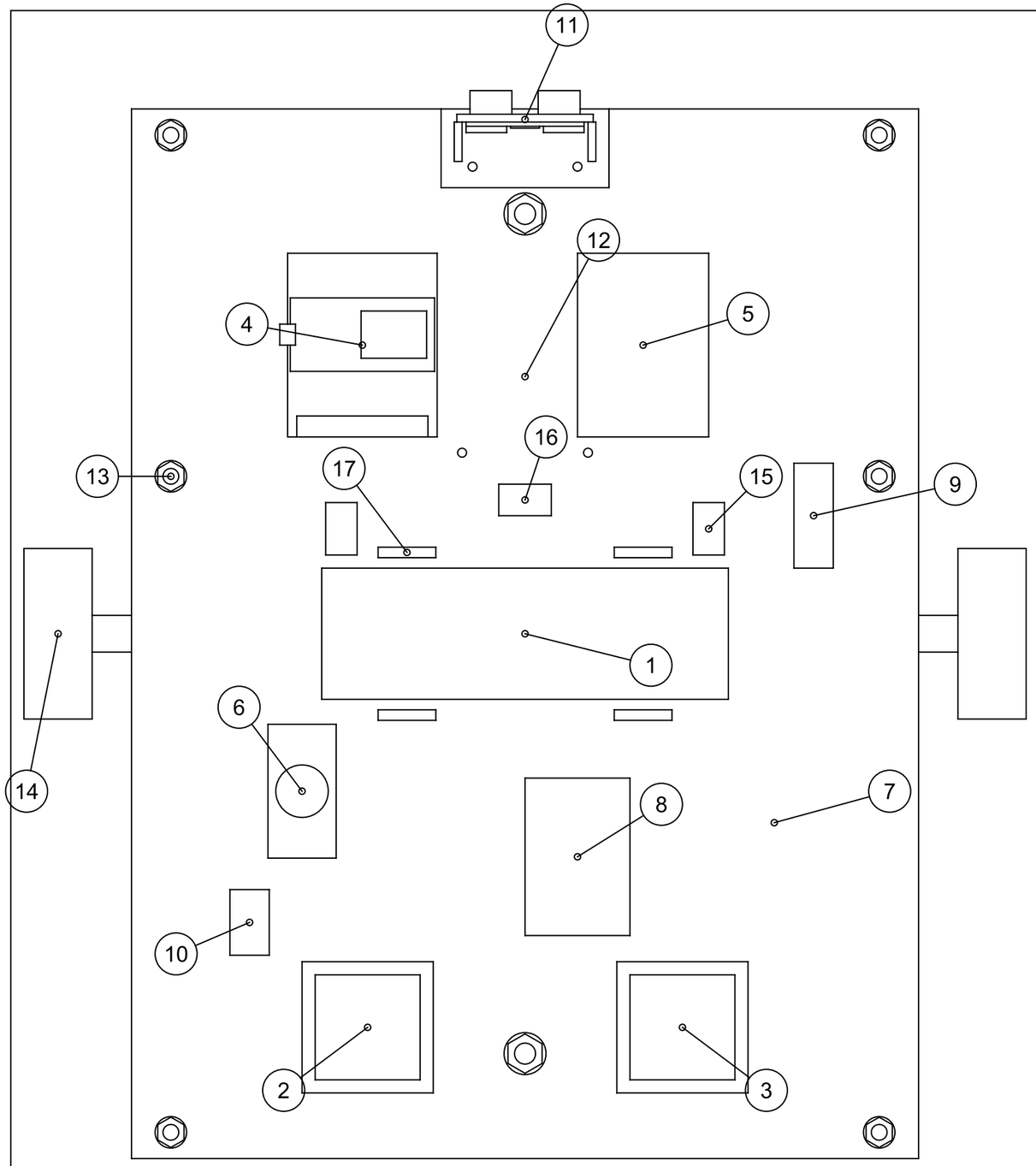


PLAN VIEW FRONT -->

NOTES

1. All dimensions in mm. Weights and loads in kg.
2. Drive: 2 x JGB37-520 12 V 110 RPM motor sets, wheels D65 x 26.
3. Casters: 2 printed ball casters (D25.4 ball) on M8 bolts (height set by nuts).
4. Frame: 6 mm acrylic plates on 6 x M6 threaded rod.
5. Bed: tipped 40 deg to the right by 2 servos on the hinge line.
6. Cargo plate on 4 printed risers; packages up to 3 kg (not weighed).
7. ESP32 (no I/O expander), 3 IR line sensors (under), HC-SR04 (front).
8. Sheets: 01 GA, 02 layout, 03-06 plates, 07 bed, 08-09 prints, 10 BOM.

PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE: GENERAL ARRANGEMENT				DWG No.: CR-V2-001	
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: 1:5	SHEET: 1 / 10	UNITS: mm	PROJECTION: 1st ANGLE	DATE: 12/11/2026	DRAWN:



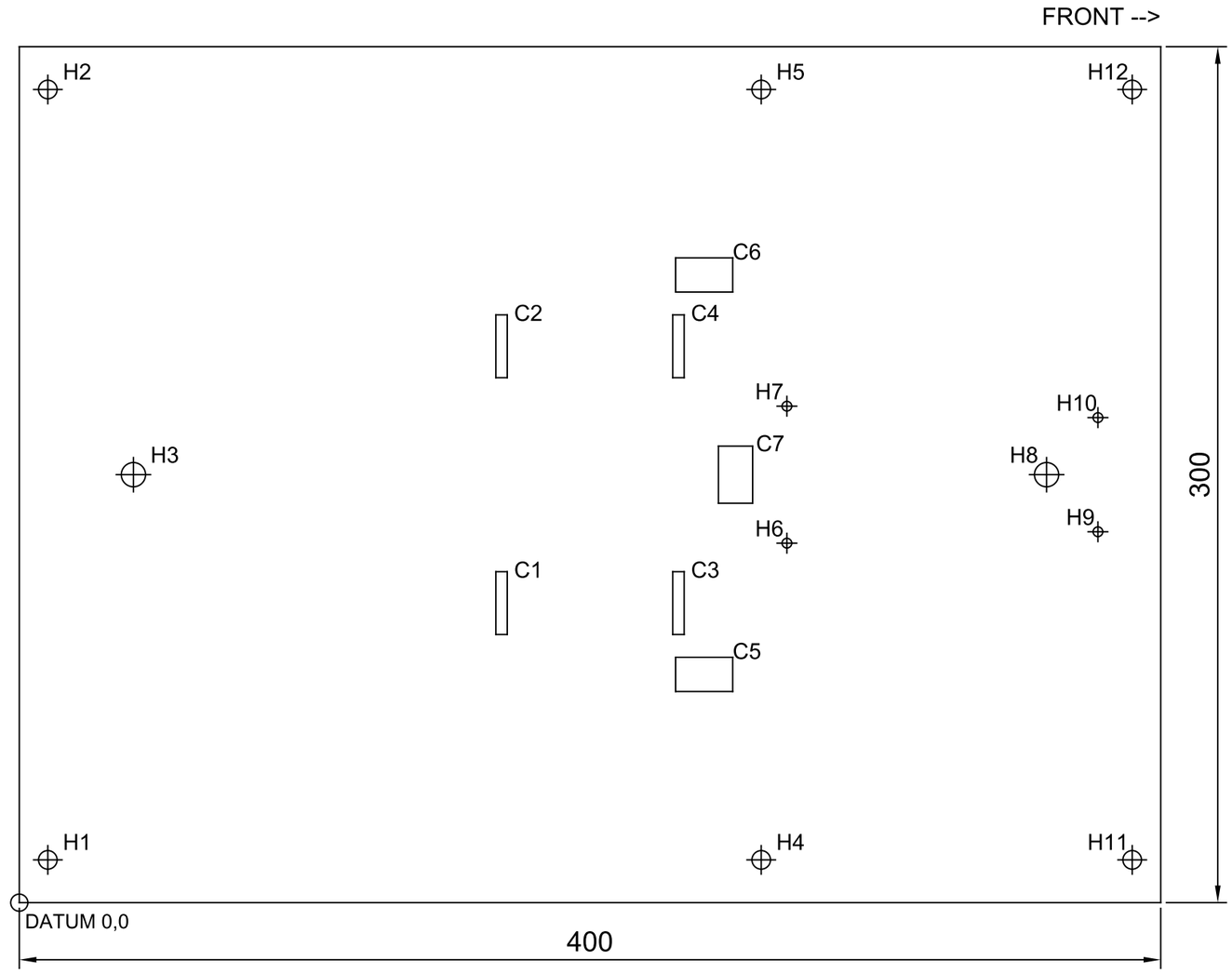
PLAN VIEW - DECK AND BED REMOVED (front of robot at the top)

No.	COMPONENT
1	3S LiPo battery 5000 mAh (lies across, slides out sideways)
2	BTS7960 motor driver - left motor
3	BTS7960 motor driver - right motor
4	ESP32 DevKit on a screw-terminal breakout (USB to the left)
5	I/O board: resistors and dividers (perfboard)
6	Buck converter 12 V -> 5.1 V (ESP32, sensors)
7	Buck converter 12 V -> 7.4 V (bed servos)
8	WAGO power buses (+12 V main / motor / 5 V / GND)
9	Inline fuse 10 A
10	Buzzer
11	HC-SR04 ultrasonic on printed bracket
12	3 IR line sensors on a printed bar (under the plate)
13	M6 threaded rod (x6)
14	Drive wheel D65 (motor under the plate)
15	Motor / encoder cable slot (x2)
16	Line sensor cable slot
17	Battery strap slots (x4)

NOTES

Small boards are held with VHB tape.
Keep 10 cm slack on every wire that goes up to the deck or the bed.
Wiring: see BUILD_GUIDE.md weeks 2 and 3, and the schematics E01-E04.

PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE: LAYOUT - DECK REMOVED				DWG No.: CR-V2-002	
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: 1:2	SHEET: 2 / 10	UNITS: mm	PROJECTION: 1st ANGLE	DATE: 12/11/2026	DRAWN:



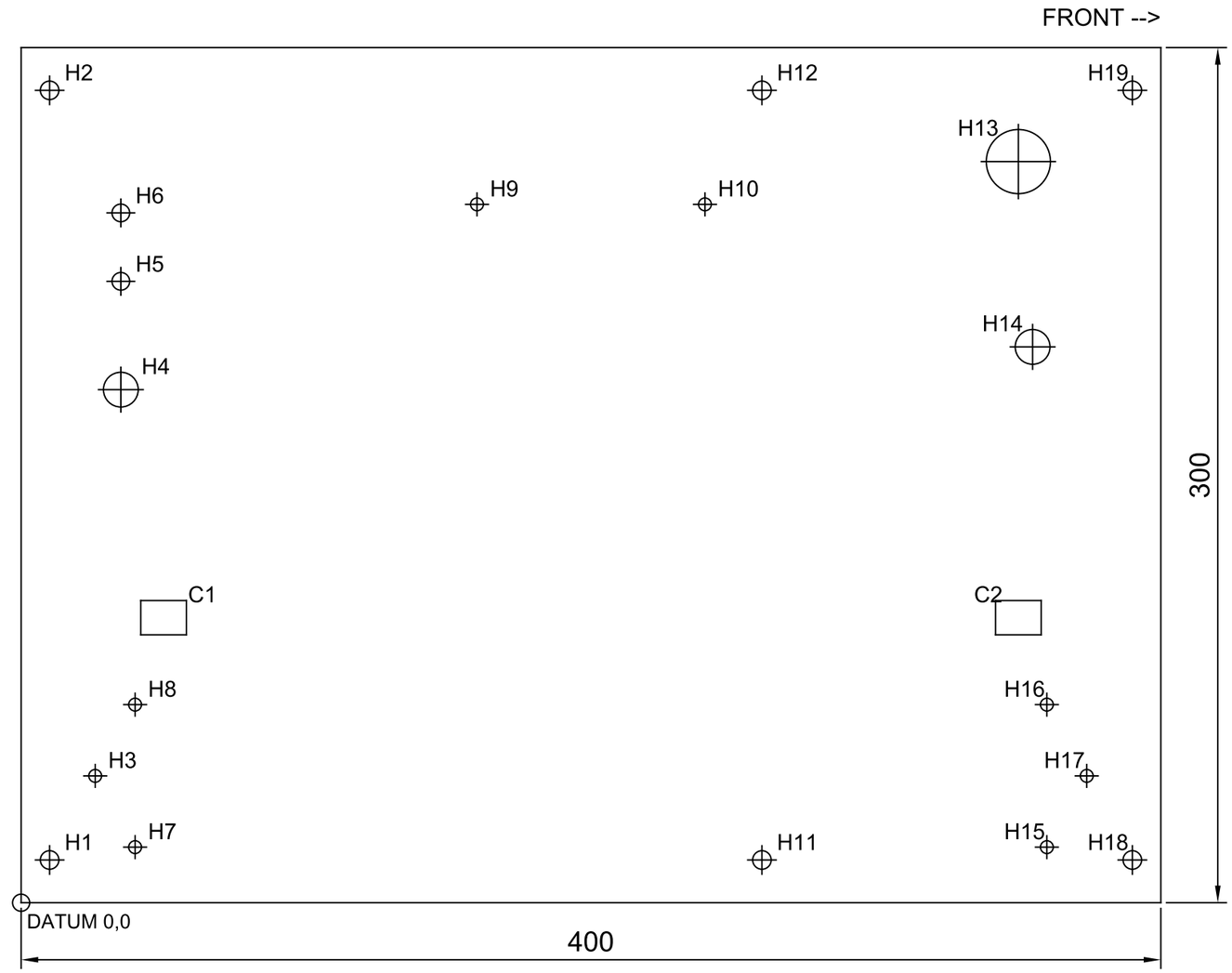
ID	X	Y	SIZE	FOR
H1	10.0	15.0	D6.5	M6 threaded rod
H2	10.0	285.0	D6.5	M6 threaded rod
H3	40.0	150.0	D8.5	ball caster M8 bolt
H4	260.0	15.0	D6.5	M6 threaded rod
H5	260.0	285.0	D6.5	M6 threaded rod
H6	269.0	126.0	D3.4	M3 sensor brackets
H7	269.0	174.0	D3.4	M3 sensor brackets
H8	360.0	150.0	D8.5	ball caster M8 bolt
H9	378.0	130.0	D3.4	M3 sensor brackets
H10	378.0	170.0	D3.4	M3 sensor brackets
H11	390.0	15.0	D6.5	M6 threaded rod
H12	390.0	285.0	D6.5	M6 threaded rod
C1	169.0	105.0	4 x 22	battery strap
C2	169.0	195.0	4 x 22	battery strap
C3	231.0	105.0	4 x 22	battery strap
C4	231.0	195.0	4 x 22	battery strap
C5	240.0	80.0	20 x 12	cables (motors, line sensors)
C6	240.0	220.0	20 x 12	cables (motors, line sensors)
C7	251.0	150.0	12 x 20	cables (motors, line sensors)

X, Y = centre of the feature from DATUM 0,0 (H = hole, S = slot, C = cut-out)

NOTES

1. Material: 6 mm acrylic, 1 off. Washers under the M3 heads and nuts.
2. Laser-cut from cad/output/dxf/bottom_plate.dxf. All dimensions in mm, +-0.2.
3. The motor brackets' holes are drilled on assembly, to suit the brackets - see BUILD_GUIDE week 1.
4. Small holes: 2 for the line-sensor bar (underneath), 2 for the ultrasonic bracket.

PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE: BOTTOM PLATE				DWG No.: CR-V2-003	
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: 1:2	SHEET: 3 / 10	UNITS: mm	PROJECTION: 1st ANGLE	DATE: 12/11/2026	DRAWN:



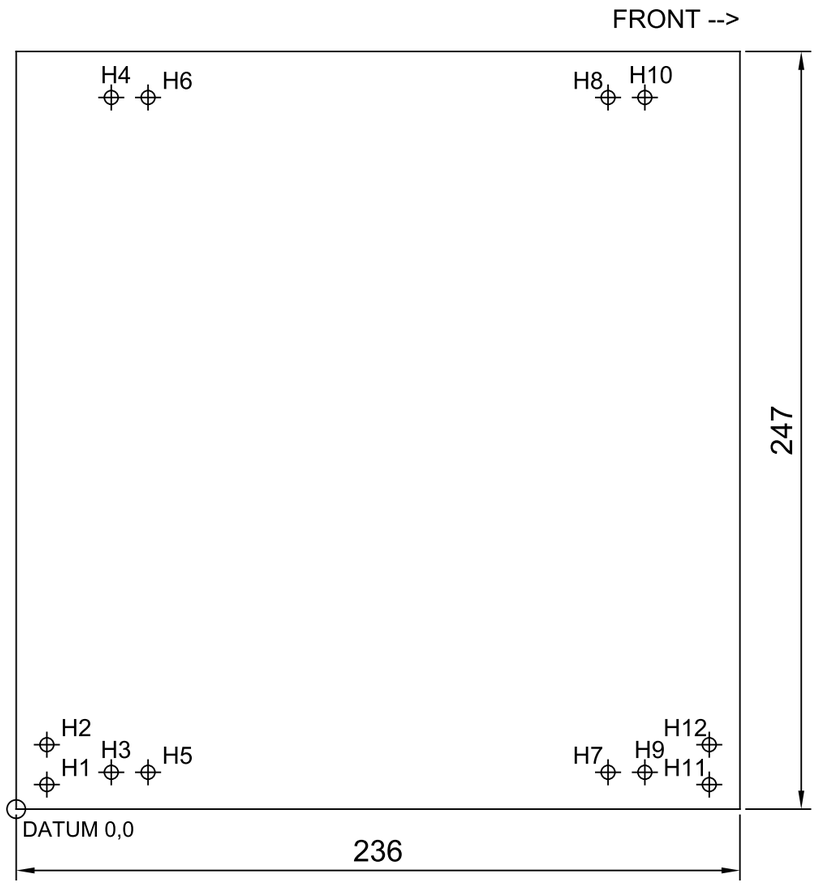
ID	X	Y	SIZE	FOR
H1	10.0	15.0	D6.5	M6 threaded rod
H2	10.0	285.0	D6.5	M6 threaded rod
H3	26.0	44.5	D4.5	M4: servo mount / bed rest
H4	35.0	180.0	D12.2	switch / start button
H5	35.0	218.0	D6.2	5 mm LED clip
H6	35.0	242.0	D6.2	5 mm LED clip
H7	40.0	19.5	D4.5	M4: servo mount / bed rest
H8	40.0	69.5	D4.5	M4: servo mount / bed rest
H9	160.0	245.0	D4.5	M4: servo mount / bed rest
H10	240.0	245.0	D4.5	M4: servo mount / bed rest
H11	260.0	15.0	D6.5	M6 threaded rod
H12	260.0	285.0	D6.5	M6 threaded rod
H13	350.0	260.0	D22.5	E-stop 22 mm
H14	355.0	195.0	D12.2	switch / start button
H15	360.0	19.5	D4.5	M4: servo mount / bed rest
H16	360.0	69.5	D4.5	M4: servo mount / bed rest
H17	374.0	44.5	D4.5	M4: servo mount / bed rest
H18	390.0	15.0	D6.5	M6 threaded rod
H19	390.0	285.0	D6.5	M6 threaded rod
C1	50.0	100.0	16 x 12	servo cable
C2	350.0	100.0	16 x 12	servo cable

X, Y = centre of the feature from DATUM 0,0 (H = hole, S = slot, C = cut-out)

NOTES

1. Material: 6 mm acrylic (keep the protective film), 1 off.
2. Laser-cut from cad/output/dxf/top_deck.dxf. All dimensions in mm, +-0.2.
3. Every hole is cut by the laser; nothing is drilled by hand.

PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE: TOP DECK				DWG No.: CR-V2-004	
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: 1:2	SHEET: 4 / 10	UNITS: mm	PROJECTION: 1st ANGLE	DATE: 12/11/2026	DRAWN:



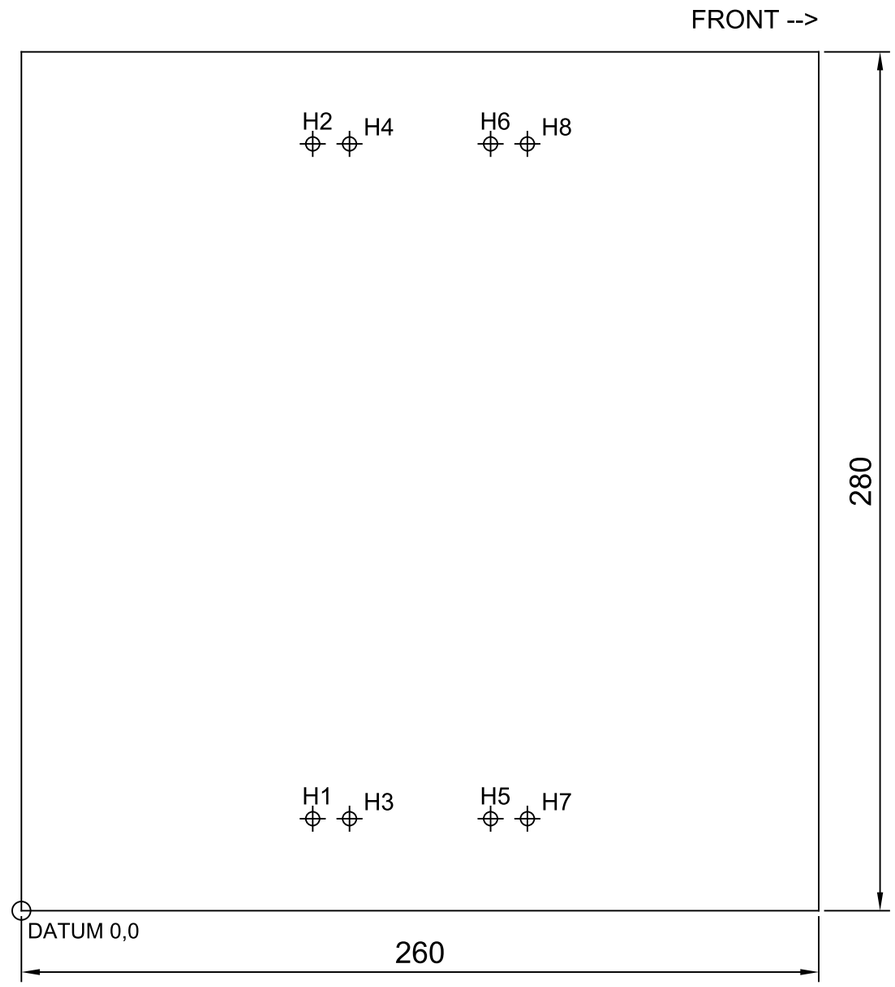
ID	X	Y	SIZE	FOR
H1	10.0	8.0	D4.5	M4: horn adapter / riser
H2	10.0	21.0	D4.5	M4: horn adapter / riser
H3	31.0	12.0	D4.5	M4: horn adapter / riser
H4	31.0	232.0	D4.5	M4: horn adapter / riser
H5	43.0	12.0	D4.5	M4: horn adapter / riser
H6	43.0	232.0	D4.5	M4: horn adapter / riser
H7	193.0	12.0	D4.5	M4: horn adapter / riser
H8	193.0	232.0	D4.5	M4: horn adapter / riser
H9	205.0	12.0	D4.5	M4: horn adapter / riser
H10	205.0	232.0	D4.5	M4: horn adapter / riser
H11	226.0	8.0	D4.5	M4: horn adapter / riser
H12	226.0	21.0	D4.5	M4: horn adapter / riser

X, Y = centre of the feature from DATUM 0,0 (H = hole, S = slot, C = cut-out)

NOTES

1. Material: 6 mm acrylic, 1 off.
2. Laser-cut from cad/output/dxf/tilt_plate.dxf. All dimensions in mm, +-0.2.
3. The bottom edge here (Y = 0) is the RIGHT edge of the robot: it lies on the hinge axis.

PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE: TILTING PLATE				DWG No.: CR-V2-005	
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: 1:2	SHEET: 5 / 10	UNITS: mm	PROJECTION: 1st ANGLE	DATE: 12/11/2026	DRAWN:



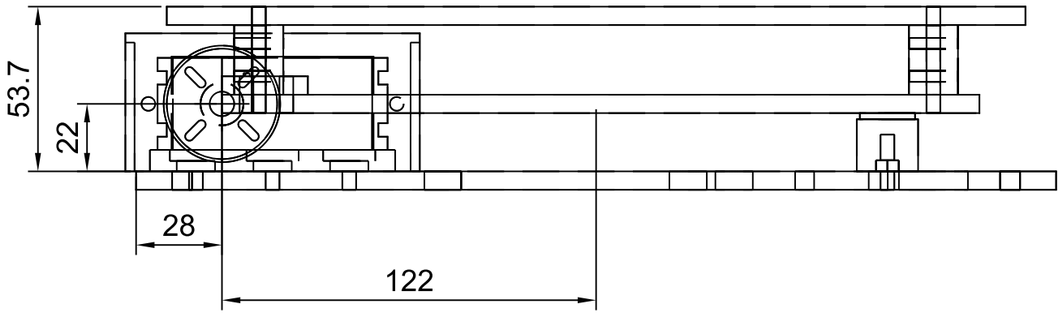
ID	X	Y	SIZE	FOR
H1	95.0	30.0	D4.5	M4 countersunk: bed risers
H2	95.0	250.0	D4.5	M4 countersunk: bed risers
H3	107.0	30.0	D4.5	M4 countersunk: bed risers
H4	107.0	250.0	D4.5	M4 countersunk: bed risers
H5	153.0	30.0	D4.5	M4 countersunk: bed risers
H6	153.0	250.0	D4.5	M4 countersunk: bed risers
H7	165.0	30.0	D4.5	M4 countersunk: bed risers
H8	165.0	250.0	D4.5	M4 countersunk: bed risers

X, Y = centre of the feature from DATUM 0,0 (H = hole, S = slot, C = cut-out)

NOTES

1. Material: 6 mm acrylic, 1 off. No anti-slip mat: the package must slide off.
2. Laser-cut from cad/output/dxf/cargo_plate.dxf. All dimensions in mm, +-0.2.
3. Countersink the 8 holes on the top face by hand (90 deg) so the M4 flat heads sit flush.

PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE: CARGO PLATE				DWG No.: CR-V2-006	
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: 1:2	SHEET: 6 / 10	UNITS: mm	PROJECTION: 1st ANGLE	DATE: 12/11/2026	DRAWN:



BED FLAT - FROM THE FRONT (front servo not shown)

HOW IT WORKS

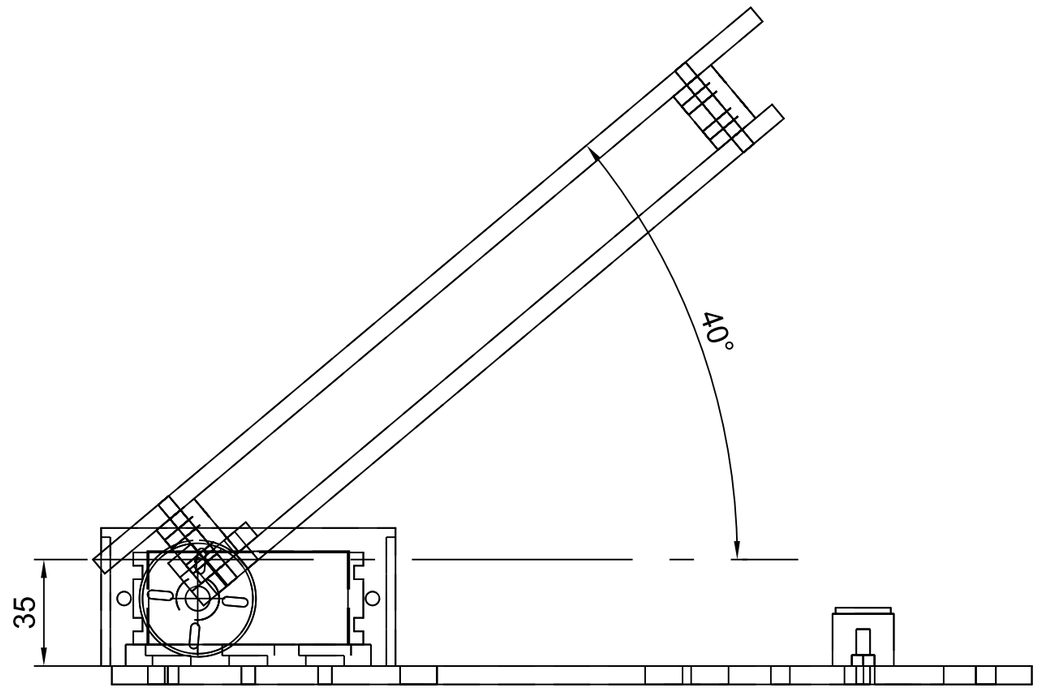
1. Two servos on the deck's right edge share one axis: their shafts ARE the hinge (front and rear end of the bed, mirror images). No hinge, screw or lever.
2. A printed horn adapter on each end of the tilting plate screws to the servo's aluminium disc horn (4 x M3). 4 printed risers carry the cargo plate.
3. Flat, the bed rests on 2 printed posts and the servos are switched off.
4. To unload, the ESP32 turns both servos slowly (20 deg/s) to 40 deg, holds 2 s, then lowers the bed and switches them off.
5. No anti-slip mat: a cardboard box slides on acrylic from about 20 deg.

TORQUE AT THE START OF THE TIP (both servos, 20 cm box)

PACKAGE	TORQUE	OF 2 x 60 kg.cm
1 kg	2.3 N.m	19 %
3 kg	4.7 N.m	40 %
5 kg	7.1 N.m	60 %

SERVOS

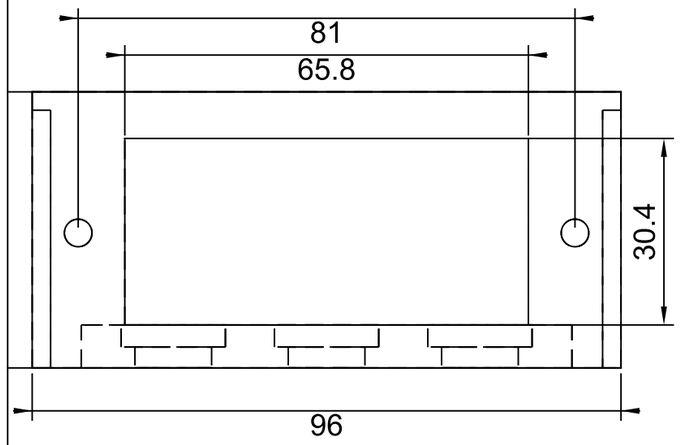
Servo DS5160: 60 kg.cm at 7.4 V, 65 x 30 x 48 mm, 75 mm over the ears, ears 31 mm above its bottom (maker's drawing). The ears are clamped, and the mount slides +-4 mm to put the shaft on the axis.
Supply: 7.4 V from its own 8 A buck converter, after the E-stop.
Signal: front servo GPIO 2, rear servo GPIO 12 (schematic E03).
Hinge axis: 188 mm above the floor, 122 mm right of the centre line.



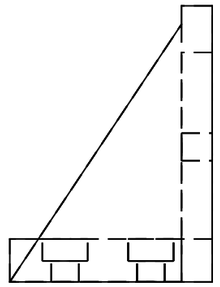
BED TIPPED 40 deg - THE PACKAGE SLIDES OFF TO THE RIGHT

(the right edge of the cargo plate ends up 6 mm outside the deck, 35 mm above it)

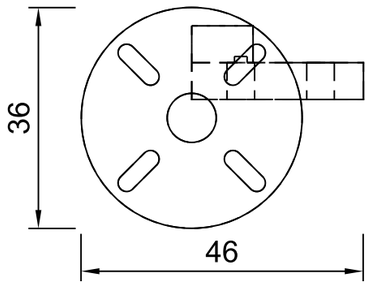
PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE: TILTING BED - SERVO HINGE				DWG No.: CR-V2-007	
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: 1:2	SHEET: 7 / 10	UNITS: mm	PROJECTION: 1st ANGLE	DATE: 12/11/2026	DRAWN:

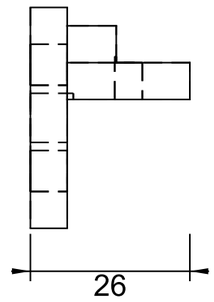


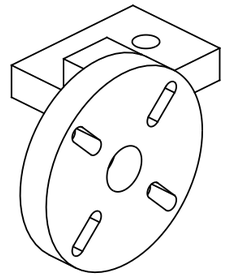
SERVO MOUNT (front), BED SIDE



SIDE



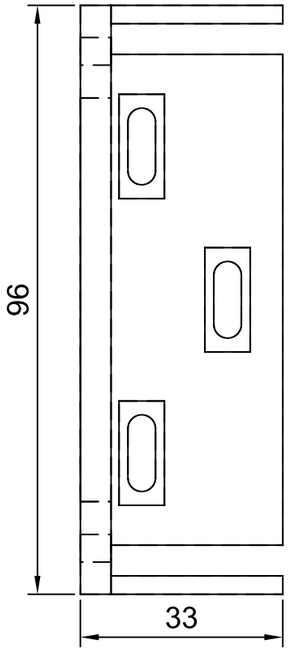




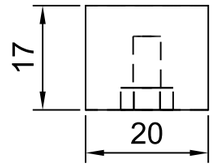
HORN ADAPTER (front) - FROM THE SERVO

4 radial slots 2.8 wide, r 9.5-15: any disc horn with a 19-30 mm screw
circle; its M3 x 8 screws cut their own thread. 2 x D4.5: M4 down through the tilting plate.

Window 65.8 x 30.4: the servo body goes through it, its ears rest on this face.
2 x D4.5: the ear clamp screws, 81 apart (M4 x 16, nut behind).
3 x slots 4.5 wide, +-4 mm along the robot: M4 x 12 (NO washer) up through the deck into nuts in the slots - slide the mount until the servo shaft is on the hinge axis, then tighten.
Rear mount = mirror image. Print lying on the ears' face: no supports.

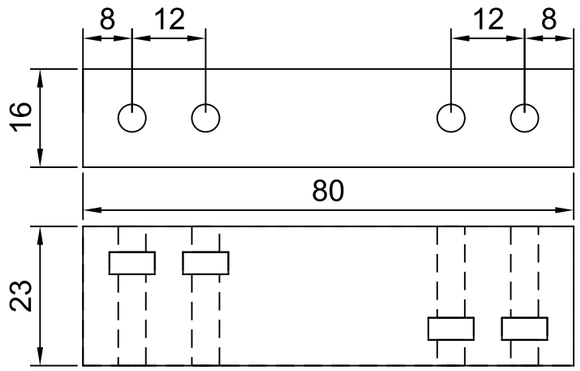


PLAN



BED REST (2 off)

M4 nut pocket in the underside; D4.5 x 12 deep.
M4 x 10 up through the deck into the nut.

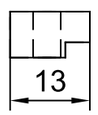


BED RISER (4 off)

4 x D4.5 through; an M4 nut slides into each 7.4 x 3.6 slot

NOTES

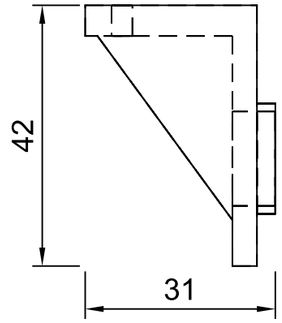
1. PETG, 0.2 mm layers, 4 walls. STLs (cad/output/parts/) lie in their printing position.
2. Servo mounts 1 + 1 (front + rear) and ear clamps 4: 40 % infill.
3. Horn adapters 1 + 1: 100 % infill, round wall on the bed, supports on the build plate only.
4. Bed rests 2, risers 4 (on the 80 x 16 face): 40 % infill.



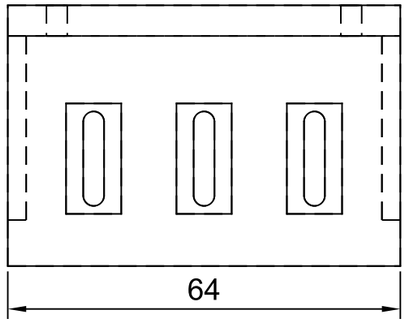
SERVO EAR CLAMP (4 off)

Its tip presses on the ear; M4 x 16 through it.

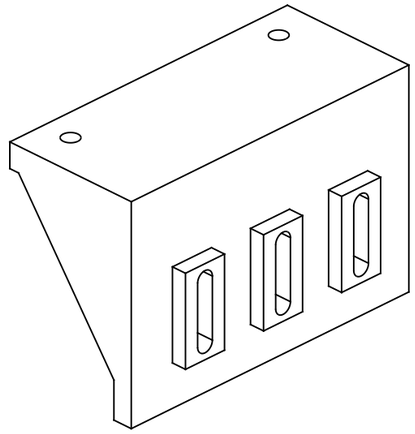
PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE: PRINTED PARTS (1/2) - BED				DWG No.: CR-V2-008	
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: 1:1	SHEET: 8 / 10	UNITS: mm	PROJECTION: 1st ANGLE	DATE: 12/11/2026	DRAWN:



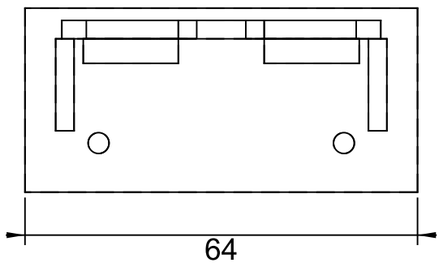
LINE SENSOR BAR - SIDE



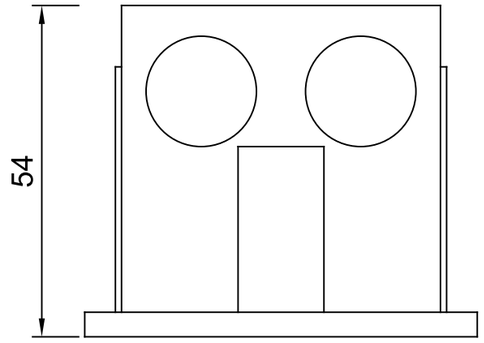
FROM THE FRONT



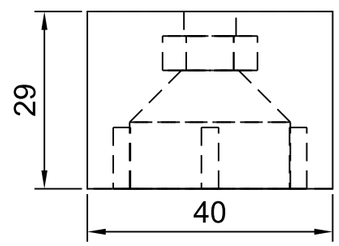
Flange under the bottom plate: 2 x D3.4, 48 mm apart (M3 x 16 + nut).
3 vertical slots 3.4 wide (12 mm of travel) on 3 mm pads, 18 mm apart: one M3 x 12 + nut per IR module. Slide each module until its LEDs are 5-10 mm above the floor.



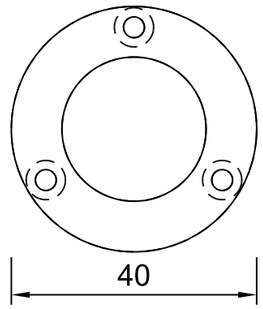
ULTRASONIC BRACKET - PLAN



FROM THE FRONT: D18 windows



BALL CASTER CUP (2 off) - SIDE



RING (2 off)

For a 25.4 mm steel ball (or a 25 mm marble). 45 deg seat: grease it.
An M8 x 50 bolt goes in from inside the cup (head in the hex pocket) BEFORE the ball; the ring then holds the ball in: 3 x M3 x 8 countersunk into the cup.

NOTES

1. PETG, 0.2 mm layers, 4 walls, 40 % infill. The STL files already lie in their printing position.
2. Line sensor bar: on the flange's top face. Ultrasonic bracket: on its base. No supports.
3. Caster cup: upside down (on its top face). Ring: flat. No supports.

PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE: PRINTED PARTS (2/2)				DWG No.: CR-V2-009	
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: 1:1	SHEET: 9 / 10	UNITS: mm	PROJECTION: 1st ANGLE	DATE: 12/11/2026	DRAWN:

No.	GROUP	DESCRIPTION	QTY	MAKE / BUY
1	Compute	ESP32 DevKitC 38-pin (ESP32-WROOM-32, USB) + 38-pin screw-terminal breakout board	1	buy (get 2)
2	Drive	JGB37-520 12 V 110 RPM gear motor SET: motor with Hall encoder (6 wires) + steel bracket + hex coupling + 65 mm wheel	2	buy
3	Drive	BTS7960 (IBT-2) 43 A motor driver	2	buy
4	Drive	Ball caster: cup (caster_cup.stl) + ring (caster_ring.stl)	2	3D print PETG
5	Drive	Steel bearing ball 25.4 mm (1 inch), or a 25 mm glass marble	2	buy (get 3)
6	Drive	M8 x 50 bolt + 2 M8 nuts + 2 M8 washers (one set per caster)	2	buy
7	Sensors	IR obstacle-avoidance module, 3 pins (VCC, GND, OUT), one blue pot (FC-51 / HW-201 type)	3	buy (get 4)
8	Sensors	Line sensor bar (line_sensor_bar.stl)	1	3D print PETG
9	Sensors	HC-SR04 ultrasonic distance sensor	1	buy (get 2)
10	Sensors	Ultrasonic sensor bracket (sonar_bracket.stl)	1	3D print PETG
11	Tilting bed	Cargo plate 260x280x6 (cargo_plate.dxf)	1	laser-cut 6 mm acrylic
12	Tilting bed	Bed riser (bed_riser.stl)	4	3D print PETG
13	Tilting bed	Tilting plate 236x247x6 (tilt_plate.dxf)	1	laser-cut 6 mm acrylic
14	Tilting bed	Servo DS5160, 60 kg.cm, metal gears, 180 deg, 6-8.4 V, 65 x 30 x 48 mm	2	buy
15	Tilting bed	Aluminium DISC servo horn for the servo's spline (25T on most DS5160), 38 mm or smaller (not the 4-arm horn: its arms hit the deck)	2	buy
16	Tilting bed	Servo mount (servo_mount_front.stl + servo_mount_rear.stl)	2	3D print PETG
17	Tilting bed	Servo ear clamp (servo_clamp.stl)	4	3D print PETG
18	Tilting bed	Horn adapter (horn_adapter_front.stl + horn_adapter_rear.stl)	2	3D print PETG 100 %
19	Tilting bed	Bed rest post (bed_rest.stl) + 18 mm self-adhesive rubber pad 2 mm	2	3D print PETG
20	Tilting bed	Buck converter XL4016 8 A, set to 7.4 V for the servos	1	buy
21	Power	3S LiPo 11.1 V 5000 mAh XT60 + balance charger + LiPo bag + voltage alarm	1	buy
22	Power	XL4015 5 A buck converter (5.1 V for the ESP32 and sensors)	1	buy
23	Power	Inline fuse holder + 10 A fuses, WAGO 221 connectors, XT60 pairs	1	buy
24	Power	Emergency stop 22 mm panel-mount, latching, NC contact block (must clamp a 6 mm panel)	1	buy
25	Power	Toggle switch 12 mm panel-mount, 12 V DC >= 15 A, thread >= 9 mm (6 mm deck)	1	buy
26	Controls	Push button 12 mm panel-mount, momentary, normally open (start), thread >= 9 mm	1	buy
27	Controls	5 mm LED green + red, with panel clips for a 6 mm panel (or hot glue), + 330 ohm resistors	1	buy
28	Controls	Active buzzer module, 3-pin, 3.3-5 V	1	buy
29	Chassis	Bottom plate 400x300x6 (bottom_plate.dxf)	1	laser-cut 6 mm acrylic
30	Chassis	Top deck 400x300x6 (top_deck.dxf)	1	laser-cut 6 mm acrylic
31	Chassis	M6 threaded rod, cut to 130 mm	6	cut from 1 m rod
32	Chassis	M6 nut + M6 washer	24	buy
33	Hardware	M4: x10 (bed rests, 2), x12 (servo mounts, 6, NO washer), x16 (servo ear clamps, 4), x20 (horn adapters 4 + risers from below 8), x20 COUNTERSUNK (risers through the cargo plate, 8); 32 M4 nuts; 32 M4 washers	1	buy
34	Hardware	M3: x8 (servo horns, 8), x8 countersunk (caster rings, 6), x12 + nut (motor brackets 8, holes drilled by hand; IR modules 3), x16 + nut (line sensor bar 2, sonar bracket 2); 30 M3 washers	1	buy
35	Hardware	Resistors 1k x1, 2k x1, 10k x5, 2.2k x1, 100 nF x1, small perfboard; VHB tape, velcro strap, zip ties, a little grease (casters)	1	buy

PROJECT

AUTONOMOUS CARGO ROBOT - DESIGN v2

TITLE	DWG No.
BILL OF MATERIALS	CR-V2-010

ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II

SCALE	SHEET	UNITS	PROJECTION	DATE	DRAWN
-	10 / 10	mm	1st ANGLE	12/11/2026	